Technical specification template: Medium-Voltage Switchgear

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A5. Generated 2026-11-05 07:50 UTC. Package version 0.1.0.dev0.

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Document control

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1. Scope

This template defines the technical requirements for medium-voltage switchgear for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- IEC 62271 series (high-voltage switchgear and controlgear), including metal-enclosed switchgear and circuit-breakers
- IEC 61869 series (instrument transformers)
- IEC 60255 series (measuring relays and protection equipment)
- Local utility requirements [project to add]
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Prospective fault level	[To be calculated: short-circuit	NOT CALCULATED

	study pending utility, transformer and generator data]	

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
Rated voltage	[Class for the 13.8 kV system: project to select]	Project decision
Rated current	[At least the required current; busbars need at least the incomer design current]	Pending manufacturer data
Short-circuit rating (kA, duration) and peak	[To be at least the calculated MV fault level]	NOT CALCULATED: utility fault level pending
Internal arc classification	[Project to define]	Project decision
Protection and metering	[Relay, CT and VT data: project to define]	Pending

5. Technical requirements

5.1 The switchgear shall be type tested per the applicable standard; type-test reports shall be supplied.

5.2 The short-circuit rating shall exceed the prospective fault level including the peak current, evidenced by test reports.

5.3 Interlocking, earthing switches, operating mechanisms, auxiliary supplies and control voltage: [project to define].

5.4 Protection relays shall be programmable, with settings from the approved coordination study; the manufacturer shall supply device curves.

5.5 Cable terminations, partial-discharge performance and arc-fault containment: [project to define].

6. Manufacturer data required

The tenderer shall supply the datasheets and values below with its offer, each with the document name and revision. A value without a source document is not accepted. These are also tracked in the manufacturer-data register and are imported into the calculations through the data override mechanism.

Parameter id	Parameter	Datasheet	Current status
distribution.MV-SWG-A.rated_current_a	MV-SWG-A: rated_current_a	MV switchgear data sheet and IEC 62271 verification evidence	MISSING
distribution.MV-SWG-A.short_circuit_rating_ka	MV-SWG-A: short_circuit_rating_ka	MV switchgear data sheet and IEC 62271 verification evidence	MISSING
distribution.MV-SWG-A.form_of_separation	MV-SWG-A: form_of_separation	MV switchgear data sheet and IEC 62271 verification evidence	MISSING

distribution.MV-SWG-A.ip_rating	MV-SWG-A: ip_rating	MV switchgear data sheet and IEC 62271 verification evidence	MISSING
distribution.MV-SWG-A.verification_evidence	MV-SWG-A: verification_evidence	MV switchgear data sheet and IEC 62271 verification evidence	MISSING
short_circuit.utility_fault_level_mva	Utility initial short-circuit power at the MV connection	Utility written statement of maximum and minimum fault levels and X/R at the point of connection	MISSING
short_circuit.utility_x_r_ratio	Utility X/R ratio	Utility written statement of maximum and minimum fault levels and X/R at the point of connection	MISSING
distribution.MV-CB-A.breaking_capacity_ka	MV-CB-A: breaking_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.MV-CB-A.making_capacity_ka	MV-CB-A: making_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.MV-CB-A.time_current_curve_ref	MV-CB-A: time_current_curve_ref	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.MV-CB-A.settings	MV-CB-A: settings	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING

7. Inspection and quality requirements

7.1 Review of drawings, bill of materials, type-test reports and the ITP.

7.2 Inspection of busbar insulation, clearances, interlocks and wiring at hold points [project to define].

7.3 Routine tests on every unit as required by the applicable standard and this specification.

7.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

7.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

7.6 Calibration certificates for test instruments.

8. Factory acceptance test (FAT) requirements

8.1 Routine tests: power-frequency voltage withstand, main circuit resistance, mechanical operation, interlock and wiring checks, protection secondary injection.

8.2 Witnessing: [owner / independent inspector to attend].

8.3 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9. Commissioning requirements

9.1 Insulation resistance and contact resistance, circuit-breaker timing, CT and VT checks.

9.2 Protection secondary and primary injection as specified; settings verified against the approved study.

9.3 Interlock and transfer scheme tests; energisation procedure.

9.4 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

10. Submittals and documentation

- Datasheets, drawings and type-test reports.
- Inspection and test plan, FAT procedure and report.
- Installation, operation and maintenance manuals.
- Spare parts and special tools lists.
- Warranty terms [project to define].

11. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- This template has not been reviewed by an independent engineer.